

Integrate Git with Docker for source-controlled application build

07/11/2025

Project Directory:

- |-Dockerfile
- |-package.json
- |-package-lock.json
- |-index.js

Step 1: npm initialization → `npm init -y`

```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ npm init -y
Wrote to /home/1rv24mc046_jamunaj/devopsteq/program4a/package.json:
```

```
{
  "name": "program4a",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

Step 2: Now create an index.js file and Now install the express → `npm install express`

```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ nano index.js
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ cat index.js
const express=require('express');
const app=express();
const port=3000;
app.get('/',(res,req)=>{
  res.send("hello");
});
app.listen(port,()=>{
  console.log(server is running");
});
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ npm install express

added 68 packages, and audited 69 packages in 12s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step3: create Dockerfile

```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ ls
package.json
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ nano Dockerfile
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ cat Dockerfile
FROM node:18 AS build
WORKDIR /app
COPY package*.json /.
RUN npm install
COPY . .

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD["node","index.js"]
```

Step 4: Open the browser and login to your github account and create a new repository In the github home page of your account click on the profile and go to Developer Settings and create a Personal Access Token, copy and keep it somewhere as it is visible only once, Again back to terminal and initialize the git locally → git init

```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc046_jamunaj/devopsteq/program4a/.git/
```

Step 5: Add all the files which are present in that folder to the git using the command → git add .

And Perform the initial commit to the main branch

```

1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git add .
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git commit -m "first commit"
[master (root-commit) 5d8428f] first commit
636 files changed, 72884 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json

```

Step 6: Once you created the repository the repo will provide the ssh link to remotely add the git repo to

through terminal which looks like → git remote add origin [http://github.com/<user-name>/<repo-](http://github.com/<user-name>/<repo-name.git>)

[name.git](http://github.com/<user-name>/<repo-name.git>)> Now select the branch to which all the files of the folder must need to be moved then push the

files to the branch

```

1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git branch -M main
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git remote add origin https://github.com/Jamunajayanna/program4.git

1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/devopsteq/program4a$ git push -u origin main
Username for 'https://github.com': Jamunajayanna
Password for 'https://Jamunajayanna@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 12 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 725.48 KiB | 9.30 MiB/s, done.
Total 719 (delta 142), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (142/142), done.
To https://github.com/Jamunajayanna/program4.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.

```

Step 7: Now the repo looks like

Jamunajayanna / program4

github.com/Jamunajayanna/program4

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main1 Branch0 Tags

Go to file

Add fileCode

Jamunajayanna first commit5d8428f · 17 minutes ago1 Commit

node_modulesfirst commit17 minutes ago

Dockerfilefirst commit17 minutes ago

index.jsfirst commit17 minutes ago

package-lock.jsonfirst commit17 minutes ago

package.jsonfirst commit17 minutes ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

About

No description, website, or topics provided.

Activity0 stars0 watching0 forks

Releases

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No packages published
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Languages

JavaScript52.1%

Dockerfile47.9%

Suggested workflows

Based on your tech stack

Docker image

Configure

Build a Docker image to deploy, run, or push to a registry.